Applied AI

Lecture 12

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Agenda

- Coursework
- Review History, developments
- Agentic AI
- Issues and benefits
- Future
- UK Tech sector

Coursework

Due 7th Jan 2026, 1pm.

Individual work.

Domain means area of application. Technique must be different from FYP. Suggested areas :

- *Market analysis, algorithmic trading, personal portfolio management, Education, Games, Robotics, Hospitals and medicine, Human resources and computing, Transportation, Chatbots, News publishing and writing, Marketing, Music recognition and composition, Speech and text recognition, Data mining, E-mail and spam filtering, Gesture recognition, Voice recognition, Scheduling, Traffic control, Robot navigation, Obstacle avoidance, Object recognition.*

If what you are interested in is not listed then write to me and we can add it.

Part A: Application area review

- Choose one of the problem domains from below:
- *Market analysis, algorithmic trading, personal portfolio management, Education, Games, Robotics, Hospitals and medicine, Human resources and computing, Transportation, Chatbots, News publishing and writing, Marketing, Music recognition and composition, Speech and text recognition, Data mining, E-mail and spam filtering, Gesture recognition, Voice recognition, Scheduling, Traffic control, Robot navigation, Obstacle avoidance, Object recognition.*

- Different from domain and technique in FYP.
- Brief literature review
  - How has AI been applied to this problem domain.
  - How is the task modeled? (using search, logic, ml etc)
  - Look at the notebook(ipynb) files in the tutorials and in the aima-python directory for examples.
  - You should write everything inside the notebook file.
  - If you want to, you can go over in terms of length, but try to stay focused and within 20% of the word length.

Compare and evaluate AI techniques

- During the literature review you should have identified three different AI techniques to apply to your domain.
  - These can vary from algorithms (A*, DFS,BFS) through to the application of different techniques, NN, Clustering, etc, or combination of both.
  - Note that difficulty and depth will be a factor in your final mark.
- Think about
  - What real world problems does the system try to solve?
  - What are the inputs outputs?
  - What are the pros and cons of different techniques.
  - Please don’t try to come up with a new approach.

    (contact me if you want to do a PhD)

Implementation/exploration

- This is not like your final year project but more like the notebook files. You only need to provide an example showing the application of the techniques to a new domain and data and demonstrate critical thinking in evaluating your work. Show all references.
  - A. Describe the input required for your implementation/exploration of your chosen techniques. [10]
  - B. Draw a high-level diagram describing how all the elements of your project work together to achieve the desired result. [5] (examples on next slide)
  - C. Implement a prototype. The prototype should provide a sufficiently detailed example suitable for evaluation of the results. The implementation must be done as a Jupiter notebook. [40]
- Again, difficulty will be a factor in your overall mark.

Coursework: High level diagram examples

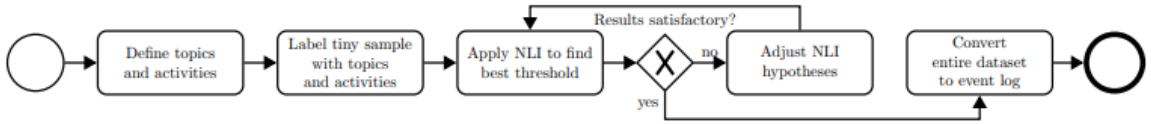
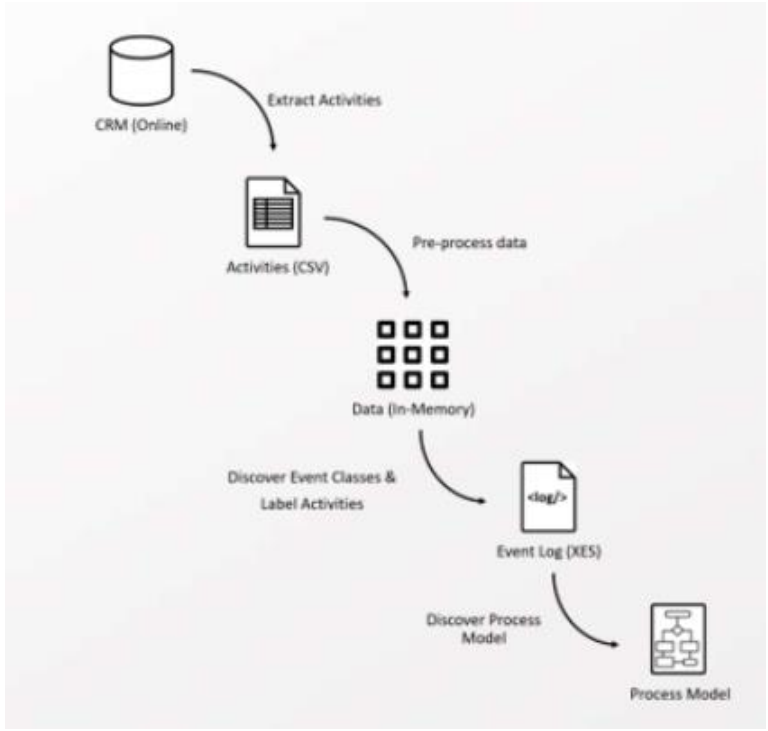


Fig. 1. Overview of the Proposed Approach



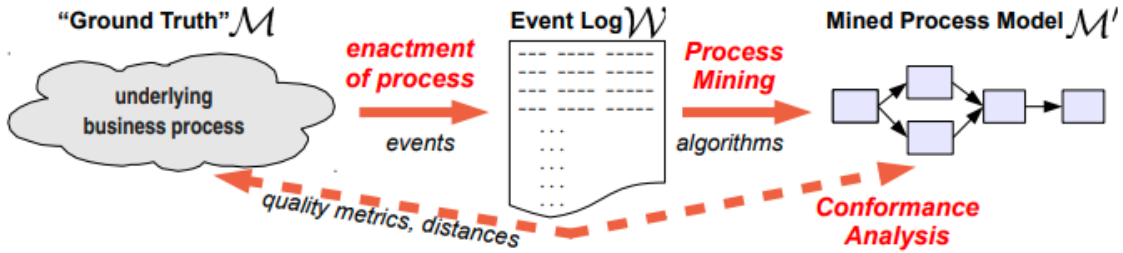


Figure 3.2: Standard View of Process Discovery.

Software Testing and Evalution (Part D)

- Part D, test the software correctness.
- Compare expected/correct results with those produced by your implementation
- How did it perform on different sizes of input? (Time/space complexity)
- Was there any unexpected behaviour and how did you deal with it. For example, loop in the graph, time issues due to exponential performance.
- Document any errors and how you would go about solving them if you had more time.

Software Testing and Evalution (Part D)

- Part D, run experiments with your prototype and interpret results.
- How do you interpret your output. How useful are your results. Do you have an evaluation metric for success. Eg, shortest path, percentage classified correctly, etc.
- What are the pros and cons of your technique/s. Are there cases that it cannot deal with?
- What is the Time space Complexity? Is it tractable for large inputs.
- Were there any surprises?
- Again, if you want to, you can exceed the word limit.

Revisit the question, What is AI?



A branch of computer science dealing with the simulation of intelligent behavior in computers.



The capability of a machine to imitate intelligent human behavior.



Strong AI: Systems that think like humans (general intelligence)



Weak AI: Just get systems to work without figuring out how human reasoning works (specific intelligence)



Use human reasoning as a model but not necessarily the end goal



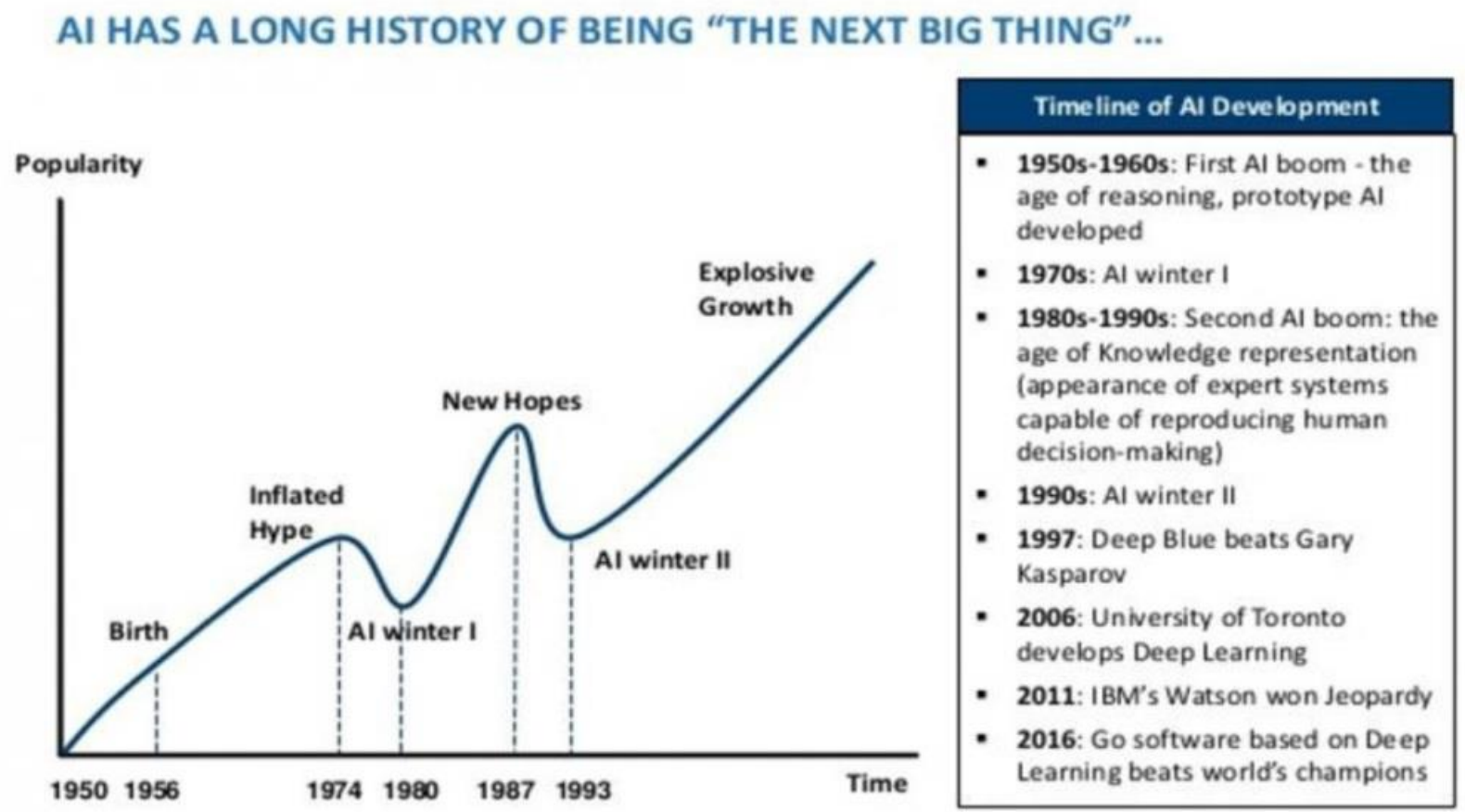
“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.”

Current AI focus 2024

- Generative AI - LLMs
  - Broad knowledge/Shallow reasoning
  - Need to improve reasoning and reliability.
  - Hybrid approaches: integration with other AI techniques
- Agentic AI
  - Autonomy – perform complex tasks independently
  - Make decisions, Learn and improve, work together.
  - Langchain/Langgraph



AI Promises


AI HAS A LONG HISTORY OF BEING "THE NEXT BIG THING"...
Popularity
Explosive Growth
New Hopes
Inflated Hype
AI winter II
Birth
AI winter I
1950
1956
1974
1980
1987
1993
Time
Timeline of AI Development
• 1950s-1960s: First AI boom - the age of reasoning, prototype AI developed
• 1970s: AI winter I
• 1980s-1990s: Second AI boom: the age of Knowledge representation (appearance of expert systems capable of reproducing human decision-making)
• 1990s: AI winter II
• 1997: Deep Blue beats Gary Kasparov
• 2006: University of Toronto develops Deep Learning
• 2011: IBM's Watson won Jeopardy
• 2016: Go software based on Deep Learning beats world's champions

Long history of reflection - The Ligthill report

Paper: "Artificial Intelligence: A General Survey" by James Lighthill, published in *Artificial Intelligence: a paper symposium* in 1973

Formed the basis for the decision by the British government to end support for AI research in most British universities.

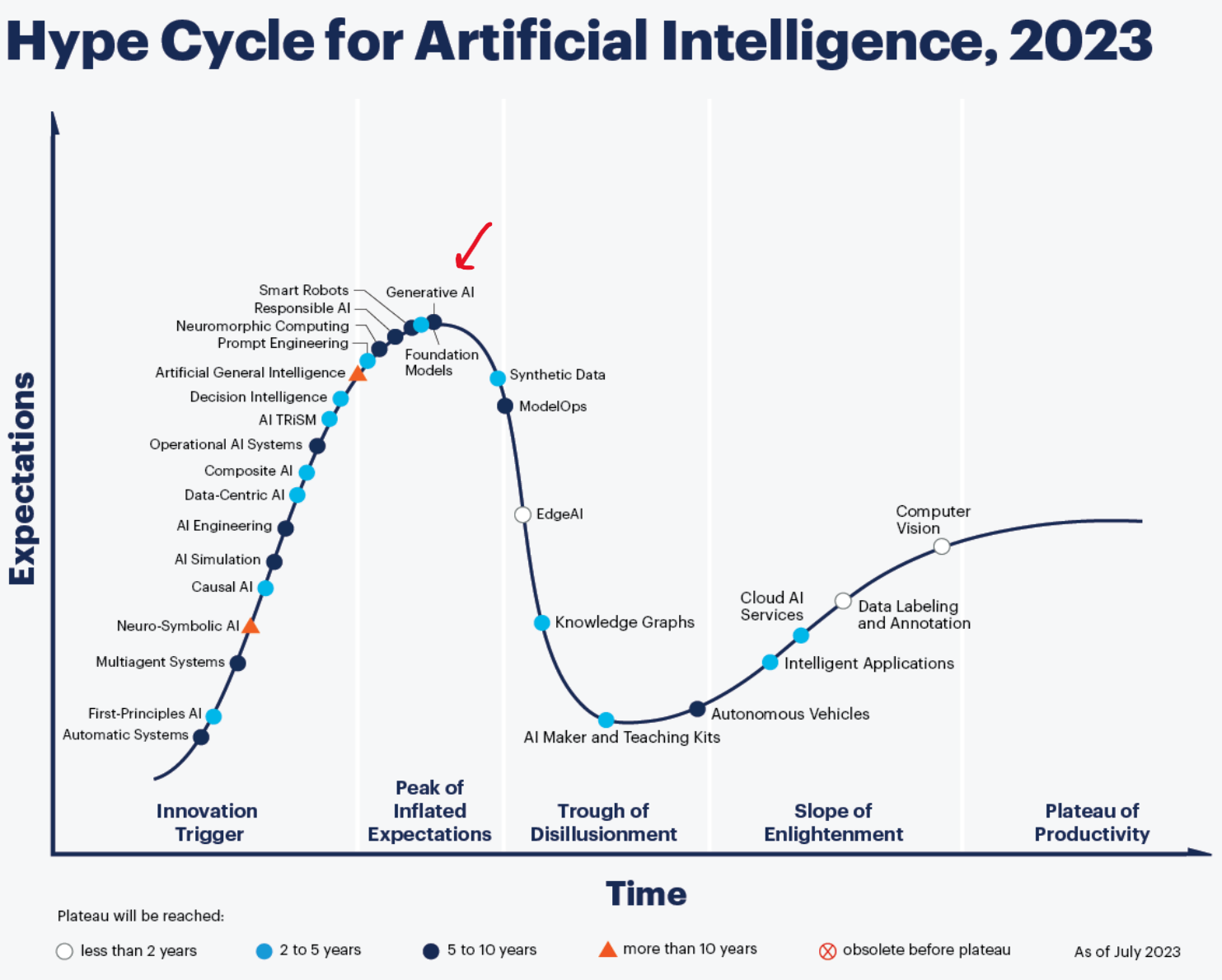
Critical of basic research in foundational areas such as robotics and language processing. Success comes from control theory, mathematics, algorithms.

AI failed at addressing combinatorial explosion when scaling up to solving “real-world” problems. (exponential complexity problem)

Are any of the statements still relevant today?

Golden Age or another AI Winter?

- Still have hype but something is different (non toy-domain results)
- Some of the success can be attributed to advancement of computational capability and availability of ‘big data’
- AI is always the next big thing, hence never attainable. Should we call it Computational Intelligence to tone down the hype?
- Money and marketing vs Scientific reality. Gartner predicts 16Trillion industry by 2030!



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Hype Cycle for Artificial Intelligence, 2024



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Issues and concerns with AI

- Explainability – can we trust it, recently discovered GPT4o1 models deceive
- Ethical/Legal implications – if AI is mainstream who is responsible?
- Inherent bias in data? AI favours men for jobs due to historic training data
- Tracking you and targeted advertising. Products can help but also target vulnerabilities.
- Displacing employees. Taxi drivers -> Self driving cars.
- Can AI achieve Consciousness? Is General AI even possible? Mind transfer? What happens if/when we reach the **singularity** – many have weighed in on this topic including – Hawkings, Musk, Kurzweil etc
- Can you think of others?



Case study – Google Project Maven

- 2018 Google collaboration with Department of Defense on project Maven made many employees quit.
- Project Maven, uses AI to analyse drone footage to detect people and objects.
- Thousands signed petition wanting Google ("Dont be evil") to do the right thing.
  - "Wrong to build technologies for military surveillance with potentially lethal outcomes"
- -> Google said it will not continue Project Maven.
- -> Also published AI principles to outline where the company stands on ethical AI

BUT

- Only general and vague claims.
  - They will not work on technologies to "directly facilitate injury to people". However, they also stated that they will continue to work with the military and government in other areas including search and rescue.
- What do you think?

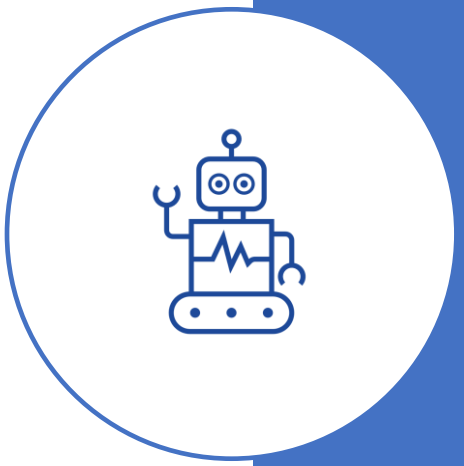
See TEDx talk: https://www.youtube.com/watch?v=cplucNW70II

Case study - Discussion

- Does this rule out googles work on Project Maven? Same technology can be used for both targeting and Search Rescue.
- Vague to the point of being meaningless. For example, for accountability and safety they state.
  - Design AI systems to be appropriately cautious.
  - Test and monitor in appropriate cases.
  - Subject them to appropriate human control.
- What does "appropriate" mean?
- But should the military overrule all surveillance even if the goal is to save lives? Positions must be clear on both sides.
- ETHICS Washing - giving the appearance of doing ethics. Regulation and compliance to avoid Ethics policing.
- Other examples: Algorithms create Filter bubbles, reinforce discrimination by excluding women from high-powered jobs, excluding certain minorities as high risk.
- **For your FYP give serious consideration to the ethical impact of your project? How your product abstracts decision making, harms users, or makes some groups worse off.**
- How is this issue being played out currently in the context of large language models like Chatgpt, Claude, and Gemini?

Advantages and benefits of AI

- Enhanced Intelligence and Decision Making: Higher intelligence brings greater insight and understanding.
  Did AlphaGo destroy the game of Go?
- Automation and Enhanced Capabilities: AI/Robots can do jobs that humans cannot. Greater efficiency 24x7, productivity can improve quality of life.
- Rapid Knowledge transfer: Copy/paste knowledge. Transfer of skills – humans need years.
- Augment Human Capabilities: Aid human research, engineering, creativity. Bicycles for the mind.
- Specialised Industry Benefits: Healthcare, smart manufacturing, customer services, process optimisation



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Text **ARTIEB757** to **07480 781235** once to join

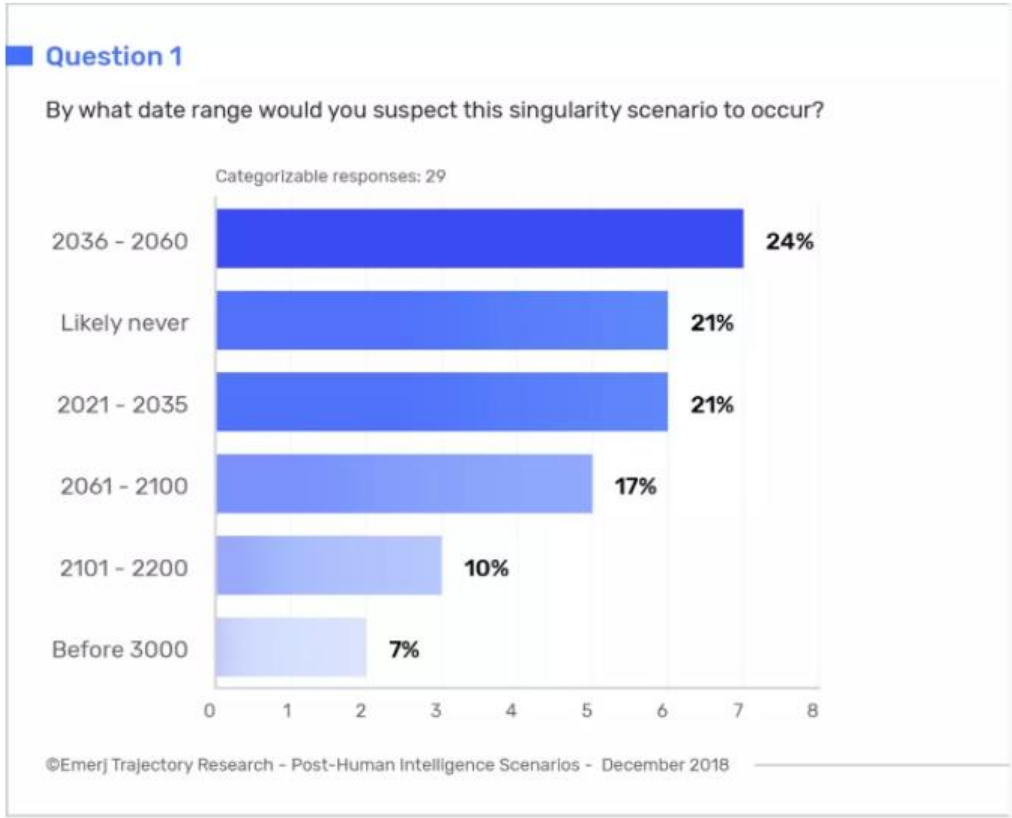
By what date range would you suspect the singularity (general AI) scenario to occur?

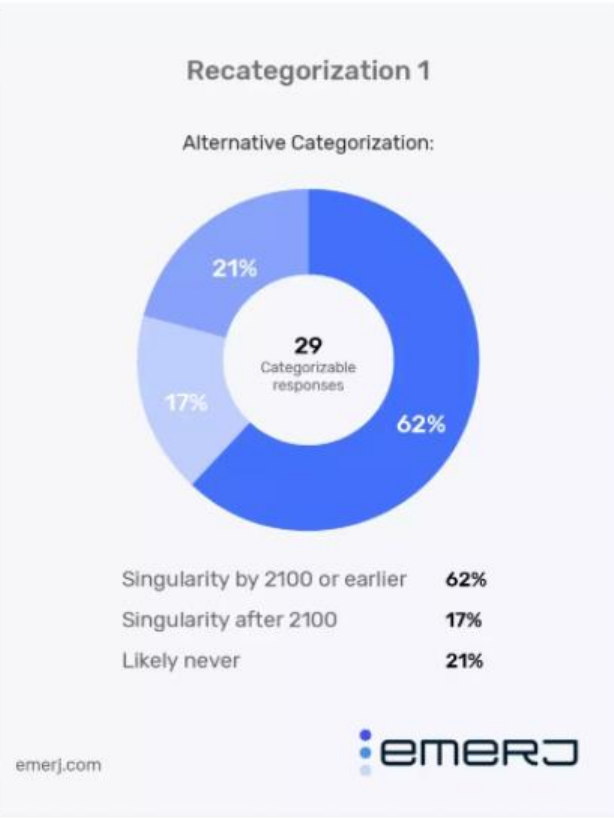
- 2021-2035
- 2036-2060
- 2061-2100
- 2101-2200
- Before 3000
- Likely never

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Predictions for AGI, Singularity 2019

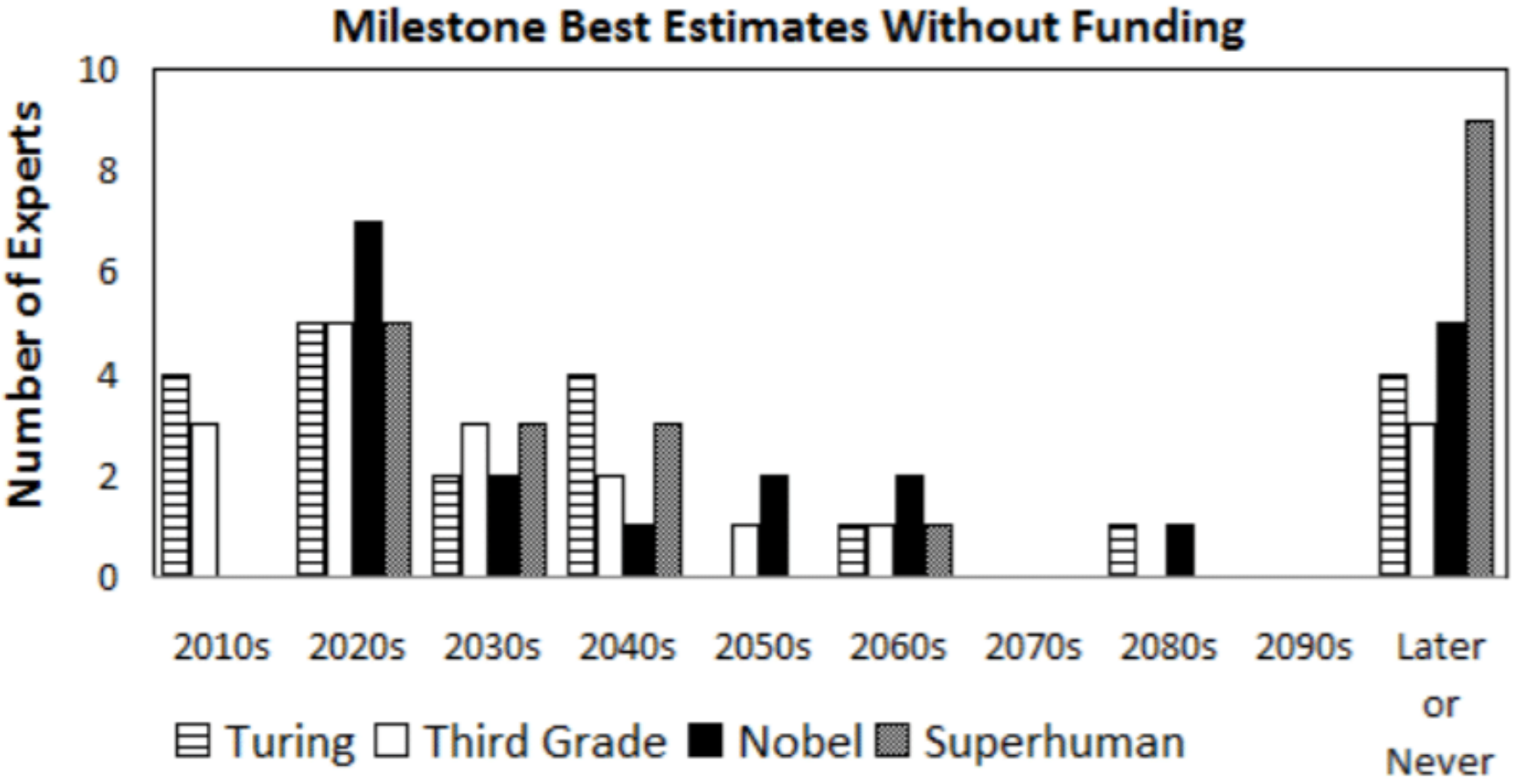
- https://emerj.com/ai-future-outlook/when-will-we-reach-the-singularity-a-timeline-consensus-from-ai-researchers/





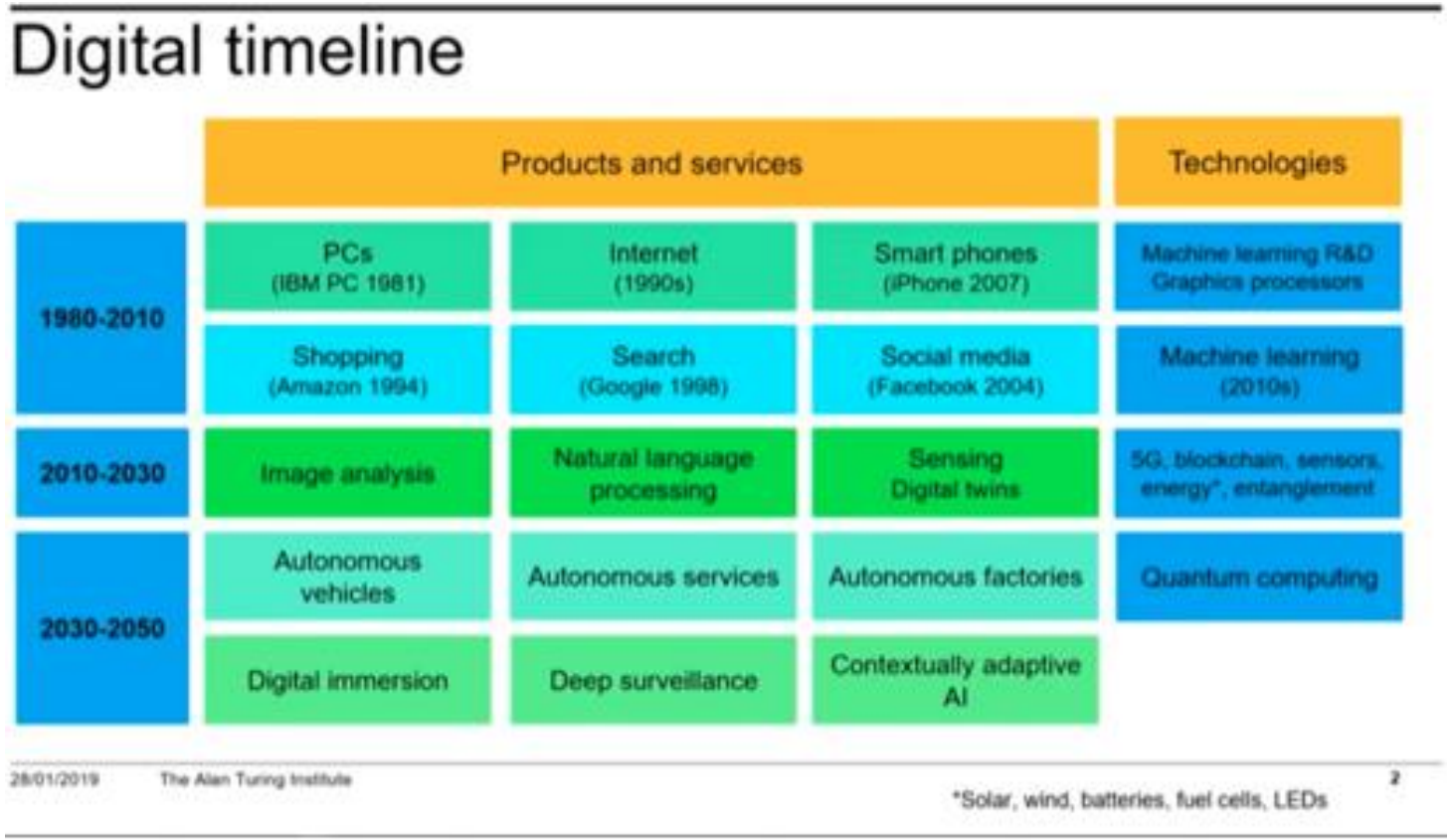
Predictions for AGI, Singularity 2023

- When will singularity happen? 1700 expert opinions of AGI [2023] (aimultiple.com)



AI Future

- Turing Lecture: Glimpsing our AI future
- https://www.turing.ac.uk/events/turing-lecture-glimpsing-our-ai-future
- Howard Covington
- “AI meets the real world”



UK Tech sector

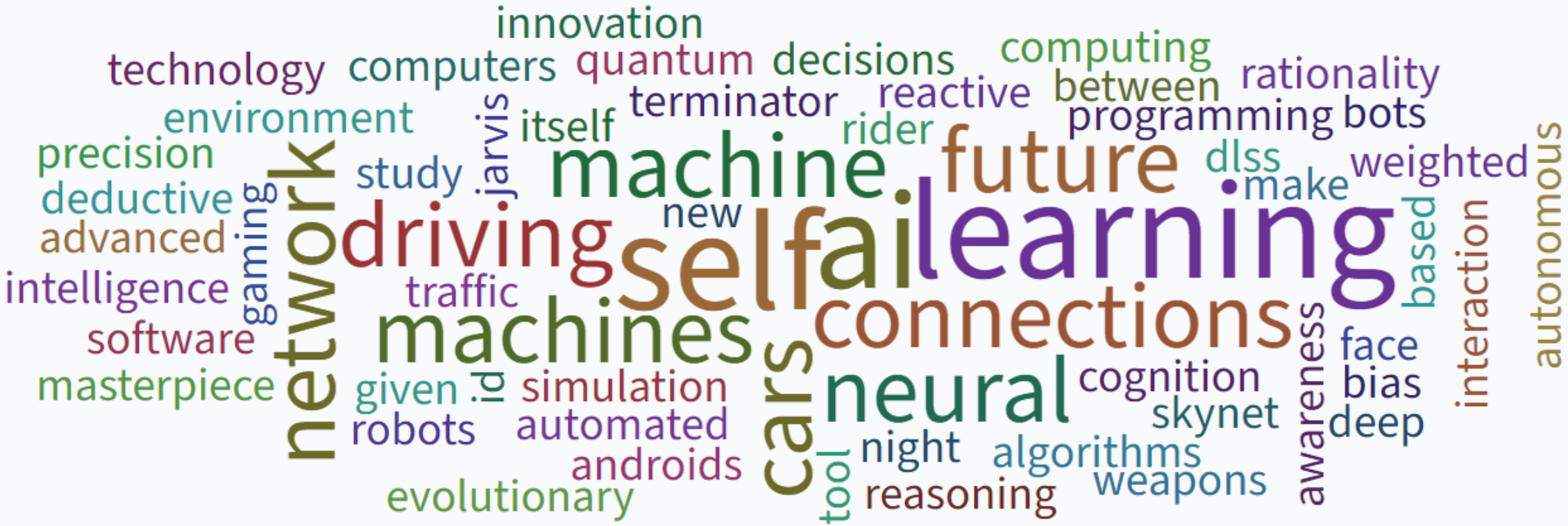
- Value of UK tech sector as of 2024 is worth £860bn
- Europe's leading tech ecosystem and
- Success stories: Ocado, Deliveroo, Monzo, Revolut
- Recent Employability Advisory Board meetings takeaway:
  - Specialist skills in high demand. AI is on the forefront.
  - Total AI-related employment increased by 29% (+14,500 jobs)

- Sector continues to grow, reasons to be optimistic!
- I look forward to inviting you to present your discoveries in future lectures as alumni.



Questions Discussion?

What words or phrases spring to mind when you think of Artificial Intelligence?



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Instructions Responses Clear responses